

Vaibhav Singh

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Relevant Research & Engineering Experience

Founder & Independent Researcher, SiliCAD

July '26 – Present

AI Agents for Analog IC Design [Demo] [Code] | Python, SKILL, MCP, Virtuoso, Eldo, Calibre

New Delhi, India

- **Spec-to-Signoff Agent:** Schematic redesign, sizing, common-centroid floorplan, symmetric routing, LVS/DRC, PEX and post-layout sim on ST 65nm, via custom SKILL plugins and an MCP server.
- **Autonomous Debug:** Traces LVS/DRC failures to root cause and repairs geometry in place (e.g., LVS short → LVS CORRECT).
- **Self-Improving Loop:** Supervisor and coding agents review runs, build new plugins, and persist lessons in an analog knowledge base.

DV CAD Intern, Xorvis Inc.

March '26 – May '26

AI-Driven Formal Verification & EDA Infrastructure | Python, LangGraph, SystemVerilog

Newark, DE, US (Remote)

- **Automated Formal Verification:** Engineered an **agentic workflow** to autonomously detect functional mismatches between natural language specifications and RTL code.
- **Semantic Extraction Pipeline:** Normalized specs & RTL into unified **Control Data Flow Graph** for semantic comparison.
- **Assertion Synthesis:** Generated targeted **SystemVerilog Assertions (SVA)** directly from identified CDFG discrepancies.

CAD Engineer (Contract), Silimate (YC S23)

Dec '25 – Feb '26

EDA Toolchain Optimization & Validation [Link] | C++, Tcl, Python, CMake

Mountain View, CA, US (Remote)

- **Liberty File Optimization:** Built a **C++ Visitor-based CCS & SI pruning engine** for Liberty (.lib) files, achieving a **75% size reduction** accelerating EDA tool loads for large-scale timing signoff and memory compiler flows.
- **Compiler Validation Pipeline:** Integrated **VlogHammer**, **Chimera**, and **Verismith** into **sv-tests**, uncovering critical edge-case failures in Verilog & SystemVerilog for EDA tools.

Undergraduate Researcher (EDA & ML), Nanoscale Devices and Circuits Lab

Jan '25 – Dec '25

MIS Modeling Extension in OpenSTA [Link] | ML, C++, Python, OpenSTA, SPICE

IIITD, Delhi, India

- **ML-Enhanced Timing Analysis:** Upgraded OpenSTA to accurately model realistic **7nm Multi-Input Switching (MIS)** behavior, recovering critical timing margins lost by standard Single-Input Switching (SIS) approximations.
- **C++ Core Integration:** Modified **OpenSTA's source code** to bypass standard liberty NLDM look-up tables, directly embedding trained **ANN/RF inference engines** into the delay calculation path.
- **Simulation Performance:** Trained models on SPICE simulation data to achieve a **10,000x speedup** with just 5% error, mathematically recovering **27.8% delay pessimism** and **3.9% slack**.

Undergraduate Engineer, Advance Multicore System Lab

Jan '25 – Aug '25

RISC-V Micro-controller for Edge AI Computing | RISC-V, SystemVerilog, FPGA, Synthesis, PD (CTS, STA)

IIITD, Delhi, India

- **Coprocessor Integration:** Integrated a custom **vector arithmetic coprocessor** with the **CV32E40X RISC-V core** using the **XIF interface**, validating functionality via RTL simulation and **FPGA deployment**.
- **RTL-to-GDSII Flow:** Executed **Logic Synthesis**, **floorplan-to-routing**, **clock tree synthesis (CTS)** and multi-stage **STA** to algorithmically resolve setup/hold violations across **PVT corners** via **gate sizing** and **buffer insertion**.

Education

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)

2022 – 2026

B.Tech in Electronics and VLSI Engineering CGPA: 7 (Core GPA: 8.5)

Delhi, India

Relevant Projects

CPU-Memory Architectural Modeling [Link] | SystemC, TLM 2.0, C++

Aug '25 – Sep '25

- **Dual-Abstraction Prototyping:** Designed a cycle-approximate **SystemC TLM 2.0** model featuring **loosely-timed** blocking loops for SW-to-CPU and **approximate-timed** non-blocking interfaces for CPU-to-Memory.

Technical Skills

Languages & HDL: C++, Python, Tcl, SKILL, SystemVerilog, Verilog, Bash, SystemC

Domain Knowledge: Analog Layout Automation, Physical Design (Floorplan, Routing), Physical Verification (DRC/LVS/PEX), STA, Timing Closure, Library Modeling (.lib, NLDM, CCS, SI), SVA, UVM, Formal Verification, Logic Synthesis, HLS

Tools & Technologies: Cadence (Virtuoso Layout XL, JasperGold, Genus, Tempus), Siemens (Calibre, Eldo), OpenSTA, Yosys, Vivado, MCP, Git, Linux, CMake

AI & Machine Learning: Agentic AI, Multi-Agent Systems, RAG, LLM, Reinforcement Learning (RL), Supervised Learning (ANN, RF, SVM)

Problem Solving: Codeforces Specialist, LeetCode (DSA) + HDLBits (Verilog) 300+ Solved